

# UniTS Anomaly Ratio Sweep

STPSat4-MRR | Generated 2026-03-12 10:53

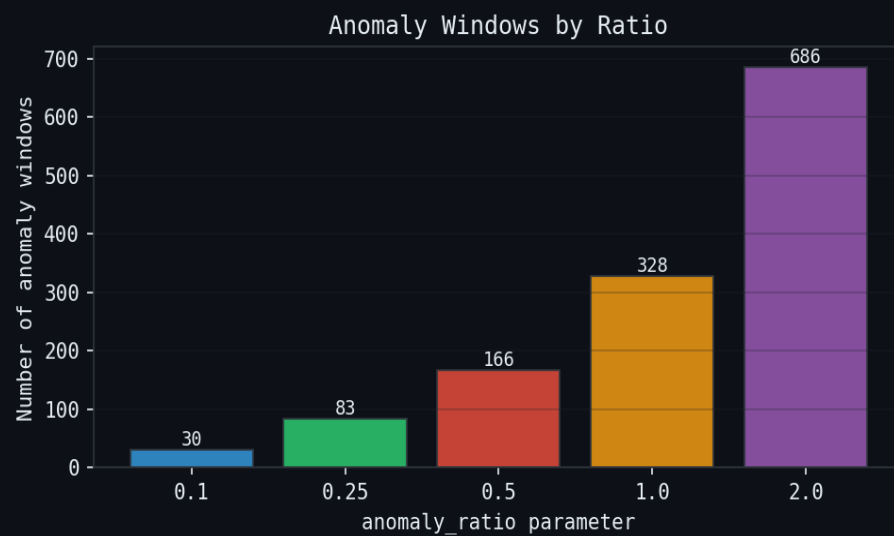
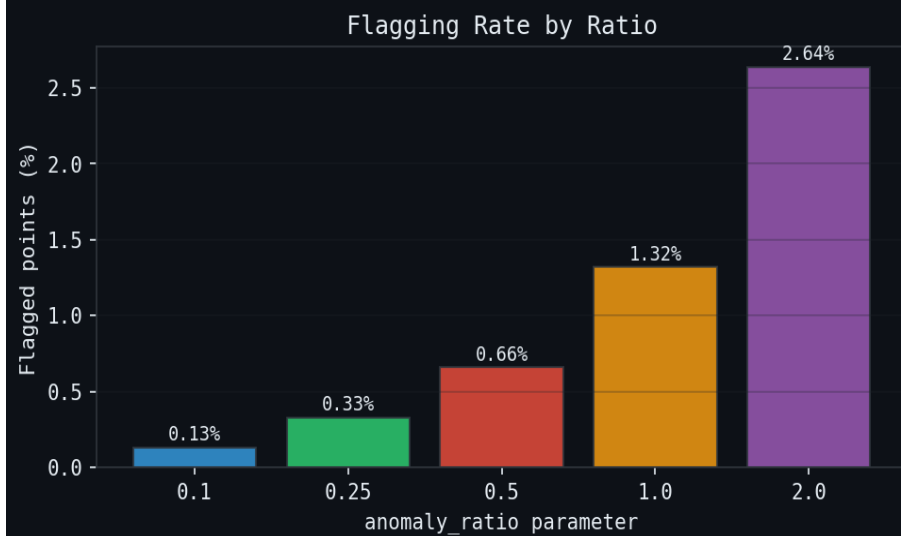
## Results by Ratio

| anomaly_ratio | Threshold | Flagged Points | Flag Rate | Windows |
|---------------|-----------|----------------|-----------|---------|
| 0.1           | 1.223264  | 57             | 0.130%    | 30      |
| 0.25          | 0.470440  | 144            | 0.330%    | 83      |
| 0.5           | 0.309005  | 288            | 0.659%    | 166     |
| 1.0           | 0.203048  | 576            | 1.319%    | 328     |
| 2.0           | 0.136119  | 1,152          | 2.637%    | 686     |

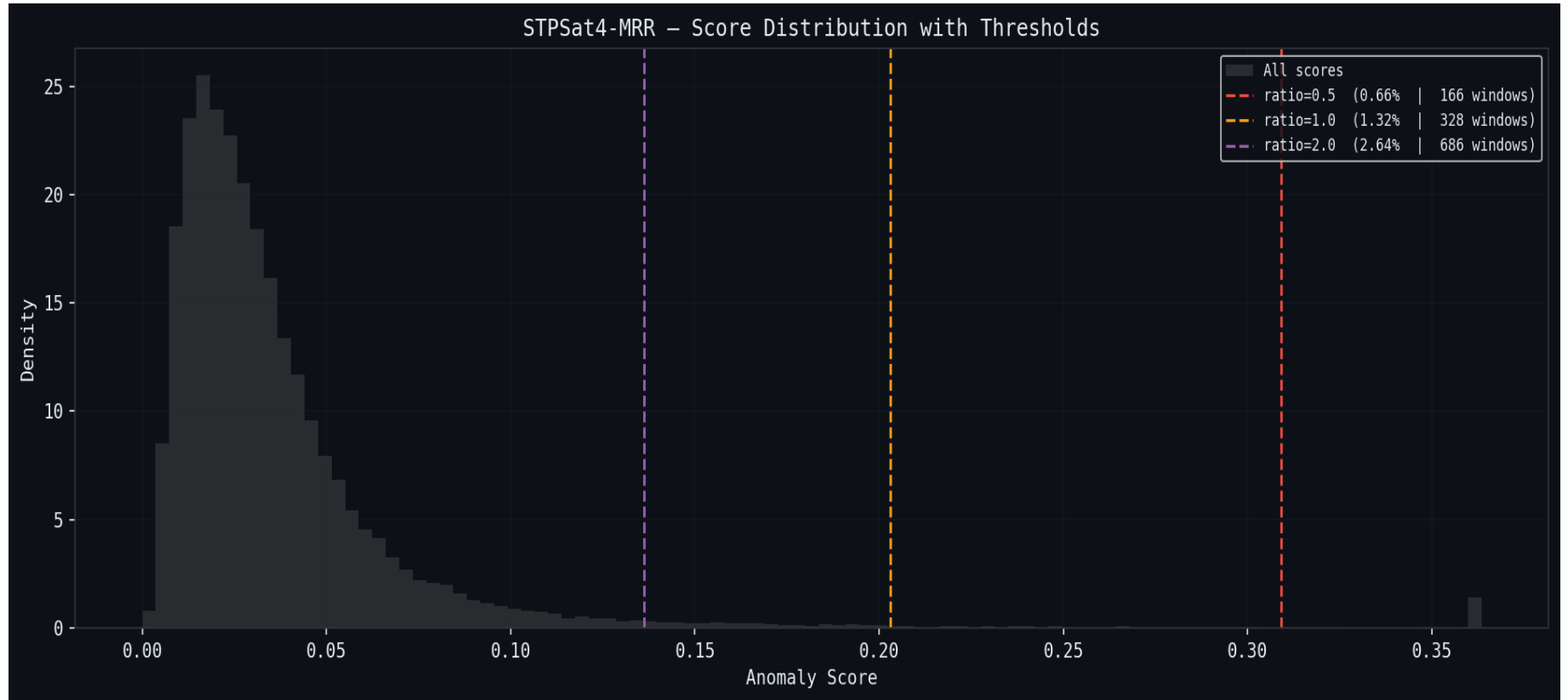
anomaly\_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

## Flagging Rate & Window Count

## STPSat4-MRR – Ratio Sweep Summary



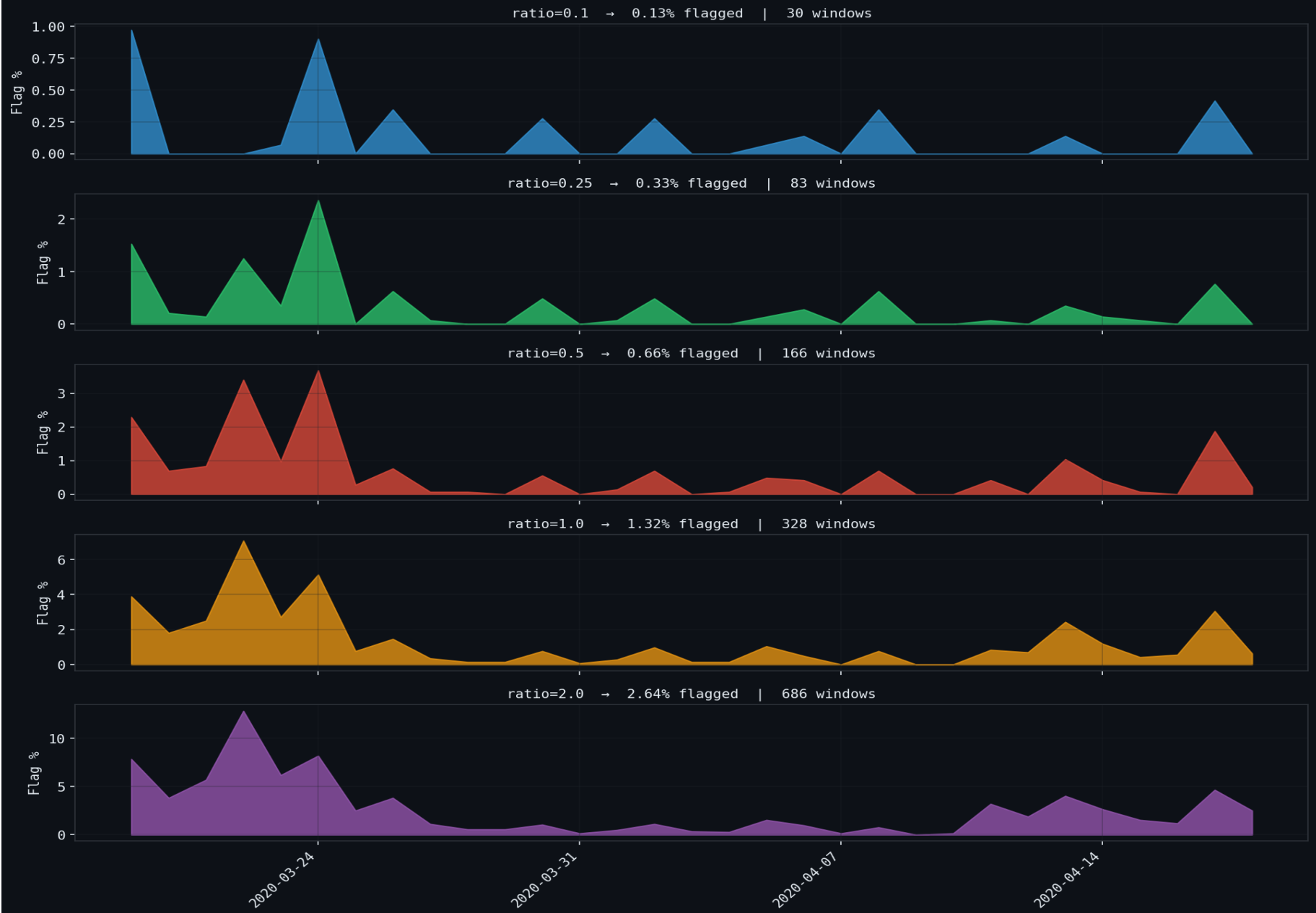
## Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

# STPSat4-MRR — Daily Flagging Rate by Ratio



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.